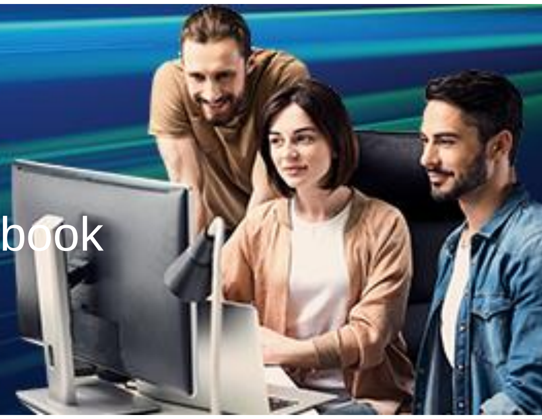
Generation Cognizant (GenC)
Learner Handbook
AWS DevOps - Basics and Advanced – v1 - Handbook



Why do we need this GenC learning Program?

Gen C learning program engages young talents with a comprehensive learning pathway, giving the millennials an opportunity to interact with Subject Matter Experts (SME), understand the corporate environment, and groom themselves.

Cognizant emphasizes on Learner Autonomy where students take charge of their own learning, with the available tools and resources. More focus is on “learning” than “teaching”. Get ready to embark your own learning adventure!

Program at a glance

Learning consisting of following skills :

- Stage 1 - AWS Basics – Fundamentals, AWS Basics - Elasticity & Management Tools
- Stage 2- AWS Cost Optimization and DNS management, AWS Advanced - Identity and Access Management and Database Services, AWS Advanced - Other AWS PaaS Services, Git & Version ControlDocker Essentials, Kubernetes Overview, Automation
- Business Aligned Lab Exercises
- Interim Evaluation And Final Evaluation

Program Highlights

- The complete learning journey is formalized using adult learning principles, where problem solving and applying the skills gained are given more importance than conceptual learning.
- Learner Autonomy is encouraged via Flipped Classroom, where the learning platform offers world class learning resources, and students would not be constrained by tutelage of an instructor.
- Get mentored by SME, whose motivation and guidance will help you accelerate in the learning journey.
- This program is applicable to Interns as well as GEN Cs.

Learning Journey with Flipped Classroom

This program encourages you to be more autonomous learners during guided self-learning hours, completing the learning objectives on your own pace and style, and get ready for the hands-on practice time.

Flipped Classroom

Self-Learning Time
- Go through the Learning Objectives
- Try to accomplish the learning objectives by accessing learning resources

Practice Time
- Get guidance from Subject Matter Expert
- Deep dive on to the learning concepts and solve a problem statement

Recommended Program Sequence

The learning journey contains Skills, followed by a Business Aligned Project.

- ➢ Stage 1 - AWS Basics – Fundamentals, AWS Basics - Elasticity & Management Tools
- ➢ Stage 2- AWS Cost Optimization and DNS management, AWS Advanced - Identity and Access Management and Database Services, AWS Advanced - Other AWS PaaS Services, Git & Version Control, Docker Essentials, Kubernetes Overview, Automation
- ➢ Business Aligned Lab Exercises
- ➢ Interim Evaluation And Final Evaluation

Stage 1 – AWS Basics

Week 1 & 2
- AWS Basics - Fundamentals
- Introduction to Cloud Computing
- Introduction to AWS
- AWS VPN Setup and Linux Basics

Week 3
- AWS Basics - Elasticity & Management Tools
- Auto scaling concepts
- Elastic Load Balancing concepts
- AWS management tools and Storage
- Stage 1 Assessment (KBA)

Stage 2 - AWS Advanced

Week 4 & 5
- AWS Cost Optimization and DNS management
- AWS Cost Explorer
- Routing Policies
- Domain Management
- Integration,Health Checks & Fail over

Week 6 & 7
- AWS Advanced - Identity and Access Management and Database Services
- Multifactor Authentication (MFA)
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Week 8
- AWS Advanced - Identity and Access
- Automation - Terrform & Python

Week 9
- **Interim Assessment (KBA) + Technical Evaluation**
- AWS Advanced - Other AWS PaaS Services
- Lambda and API Gateway

Week 10
- AWS Advanced - Other AWS PaaS Services
- Git & Version Control

Week 11
- Docker Essentials
- Kubernetes Overview

Week 12
- Kubernetes Basics & Architecture
- Kubernetes Components
- Final Assessment (KBA)+ Technical Evaluation

Key Learning Components of the Program

Cognizant has collaborated with Udemy to provide world class learning videos for the evolving future of work. These Udemy programs are woven into a learning path, empowering you to plan and learn at your style.

The program also connects you with Subject Matter Experts (SMEs) to get the professional guidance on your queries in the learning journey.

The program doesn't ONLY concentrate on the technical skilling, but also on the shaping up of the Behavioral skills. **Behavioral learning** would be done in ILT mode, with few Self-paced learning modules too.

Evaluation Model

The program continuously evaluates if you are able to apply those self-learnt skills to solve a real-time business problem. Depicted below are the four key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

- Interim Evaluation (Technical) through Video Interview
- Final Evaluation (Technical) through Video Interview

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC.

Lab Requirements

Trainees will have access to the CCL labs for the duration of their training. They can utilize the lab for sessions and hands-on practice. After using the lab, trainees are requested to shut down the lab machine to conserve / not exhaust lab access credits. If the credit limit has been exhausted, the access can't be enabled again.

Candidates need to register to the code - CTHNF099448 in Cognizant Learn and follow the below steps.
1.Open one cognizant> Cognizant learn > search for the course code.
2. Click on Register.
3. Within 5-10 minutes you will be getting new credentials Mail.
4. The Credentials mail will have the details of the CCL lab link and the User id and Password with instructions to follow.

Interim & Final Evaluation Approach

Below is the Evaluation Structure for GenC Learning Journey

The interim evaluation will be held halfway through the learning journey, while the final evaluation will take place at the end of the learning journey.

During the interim evaluation, the GenC will be interviewed by a Technical Subject Matter Expert (SME) from the Business Unit (BU) to assess your knowledge through a technical discussion.

During the final evaluation, the GenC will again be interviewed by a Technical SME from the BU to assess their knowledge through a technical discussion.

Program Completion Criteria

Stage 2 & Beyond (Advanced Skills)

Gating Criteria: Performance Health Status is Green

GenC/Intern Training	Evaluation Components	Pass Criteria	Evaluation Done By
Performance Health Status - PHS (Only for Interim and Final Topics)	Interim Technical Evaluation	Green,1Attempt	BU SME
	Final Technical Evaluation	Green,2Attempts*	

Outcome of Interim / Final evaluation will be RED, AMBER or GREEN status.

Note: 100% Completion of Hands On and Self -Learning Modules is mandatory for interim / final evaluation eligibility.

Key Check Point Intervals in the Learning Journey

Subsequent stages learning journey, your progress will be measured. On the below check point intervals, your overall Performance Health Score will be calculated as on date, and the RAG status will be arrived.

Table - Check Point Intervals

Check points	Interpreting Status
Interim Evaluation	- Green - On Track for Graduation - Red /Amber - There will not be any re-attempts given
Final Evaluation	- Green - On Track for Graduation - Red /Amber – Only 1 attempt are given (Re-attempt is not applicable if the student is Red in Interim and Final Evaluation) **Note:** If student fails after the applicable re-attempts, they will be considered as “Not Graduated”.

Icebreaker

Icebreaker session will be conducted for a duration of initial **5 days.** During the session, various topics related to Corporate Induction, Talent Management, Cognizant Agenda on Core Values, Leader Talks, Alumni, BU Mentor connects will be covered. Followed by icebreaker, technical training will kick start.

Following sessions will be covered during the **5 days** of icebreaker

- Corporate Induction
- Talent Manager Connect
- Cognizant Agenda Sessions on Core Values
- Leader Talks (Academy) and many more…

A recommended day-wise schedule is provided below for the learning, with the learning content for the day, the practice hands-on and extended hands-on to be done for the day or any other activities are listed. Few days might be interleaved to accommodate the extension due to Behavioral Training.

GenAI-Program Overview

Introduction

AI Accelerate is a comprehensive program designed to empower learners with the knowledge and skills needed to harness the transformative potential of Artificial Intelligence (AI). This handbook serves as a guide, providing essential information and resources to help learners navigate through the program successfully. From understanding the fundamentals of AI to apply the advanced concepts in real-world scenarios. AI Accelerate is your gateway to unlock the power of AI and shaping the future of technology.

Program overview.

AI Accelerate offers a learning opportunity, allowing GenCs to engage learning through the self-paced learning, Expert connect, and Knowledge assessment to measure the skills.

Focus areas

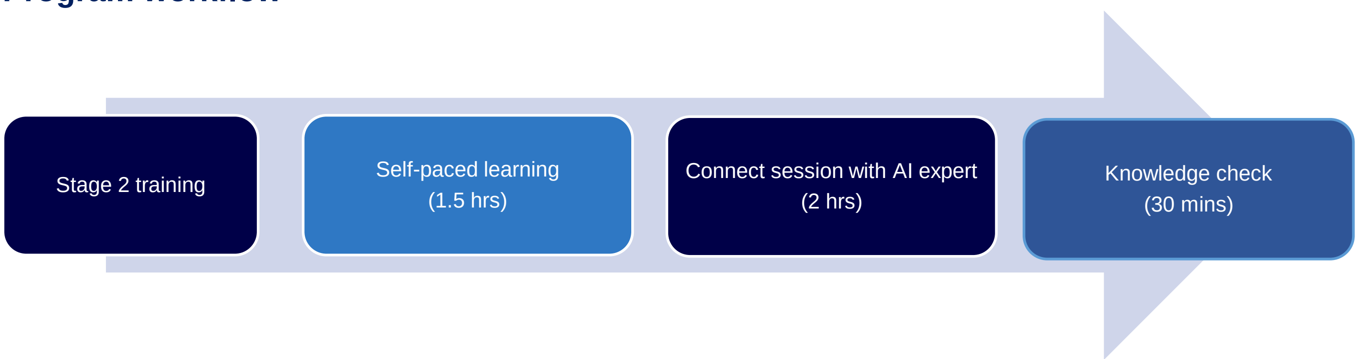
- **Learning:** GenCs will have access to curated content and resources that cover a wide range of topics related to AI, including best practices, and case studies. This learning aspect aims to deepen GenCs understanding of AI and its applications in various industries.
- **Expert connect:** GenCs will have the opportunity to connect with expert in the field of AI. This expert will provide guidance, support, and insights to help GenCs navigate their learning journey and gain valuable insights into the industry.
- **Practice sessions:** GenCs can practice the use cases provided sessions that are designed to reinforce their learning and help them apply their knowledge in real-world scenarios. These sessions will provide GenCs with hands-on experience and practical skills that are essential for success in the field of AI.

Performance outcomes

Upon completing the self-paced learning component of AI Accelerate GenCs are expected to achieve the following performance outcomes:

- GenCs will demonstrate a thorough understanding of the fundamentals of AI, including key concepts, terminology, and principles.
- GenCs will be able to apply AI concepts and techniques to solve real-world problems, demonstrating their ability to analyze, design, and implement AI solutions.
- GenCs will develop and apply critical thinking and problem-solving skills in the context of AI, enabling them to identify, analyze, and address complex challenges.
- GenCs will collaborate effectively with peers, mentors, and industry experts to achieve common goals and contribute to the advancement of AI knowledge and practice.

Program workflow



Schedule – Week

Week 1 will be focusing on AWS Fundamentals.

Udemy learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Day 1- 10

Stage 1: Days 1- 10 (Week 1& 2)

Days 1- 10 will be focusing on **AWS Basics - Fundamentals**

The following are the Scope of the training for Week 1 & 2

- Introduction to Cloud Computing
- Different types of Cloud - Public / Private / Hybrid
- Different Cloud as a Service (IaaS/PaaS/SaaS)
- Introduction to AWS
- AWS Well-Architected Framework(5 pillars): Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization.
- Amazon Virtual Private Cloud (VPC) concepts, including subnets, Route Tables, security, and networking
- VPC Peering, Direct Connect Establishment
- AWS VPN Setup
- NAT Gateway and Endpoint Connection, TGW
- Amazon EC2 Instances (On-Demand, Reserved and Spot)
- Security groups, Elastic IP, Key Pair
- Availability Zones and Regions
- EBS Volume Types, AMI / EBS Snapshots, Linux Basics.

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation.

✅ Lab 1: Launch Your First AWS Instance (Create EC2 instance, connect via SSH, attach EBS volume)

✅ Lab 2: VPC & Subnet Configuration (Create custom VPC, public/private subnets, route tables)

✅ Lab 3: VPC Peering & TGW Setup (Enable communication between VPCs)

Day 11- 15

Days 11- 15 will be focusing on **AWS Basics - Elasticity & Management Tools**

The following are the Scope of the training for Week 3

- Auto Scaling concepts(Including launch configurations, Auto Scaling groups, and scaling lifecycle)
- Elastic Load Balancing concepts(Including request routing and Load Balancer Types)
- AWS Management Tools(Including Amazon CloudWatch Rules Events and Alarms)
- AWS Storage - S3, Glacier (Including S3 Lifecycle Policy, S3 Logging)

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation.

- ✅ Lab 4: EC2 Auto Scaling & Load Balancer (Create an Auto Scaling group with ALB)
- ✅ Lab 5: S3 Storage & Lifecycle Policies (Enable versioning, lifecycle rules, and logging)

Self-Paced Learning:

Continuous Learning: Technical Enablement

Learn the AWS Basics through the below Udemy course

- Introduction to Cloud Computing on AWS for Beginners [2025]
- AWS Essentials - Hands-on Learning
- Basic AWS Architecture Best Practices - 1 Hour Crash Course
- AWS Amazon S3 Mastery Bootcamp
- AWS VPC and Networking in depth: Learn practically in 8 hrs
- Bash Scripting and Shell Programming (Linux Command Line)
- Linux for Cloud & DevOps Engineers

- [Ansible for the Absolute Beginner - Hands-On - DevOps]()
- [Chef Fundamentals: A Recipe for Automating Infrastructure]()
- [Docker for Absolute Beginners]()
- [Kubernetes for the Absolute Beginners - Hands-on]()
- [Ultimate AWS FinOps Cloud Cost Management Masterclass!]()
- [AWS VPC and Networking in depth: Learn practically in 8 hrs]()

On Day 14 , you will be taking Qualifier Mock

Stage 1 Mock Assessment

Knowledge Based Assessment on the fundamentals of AWS Basics and AWS Elasticity tools.

Note: On your **Day 16,** you will be taking Qualifier Assessment

Stage 1 Assessment(Qualifier)

Stage 1 Assessment (GenC will move on to Stage 2 training (AWS Devops Advanced) only if clear this Qualifier Assessment.

Knowledge Based Assessment on Skills AWS Basics and AWS Elasticity tools.

with passing score >=70%

Stage 2: Days 17 to 26

Day 17 to 26

Days 17 to 26 will be focusing on **AWS Cost - Optimization and DNS management**

The following are the Scope of the training for Week 4 & 5

- AWS Cost Explorer, Budgets, Billing Tags and Saving Plans
- Billing and Cost Optimization Strategies
- Route 53 - Scalable DNS & Routing
- DNS & Record Types (A, CNAME, MX, TXT, PTR)

- Routing Policies (Simple, Weighted, Latency, Failover, Geo, Multivalue)
- Domain Management (Registration, Transfers, DNS Records)
- Integration (EC2, S3, CloudFront, ELB, Global Accelerator)
- Health Checks & Failover (Active-Passive, Active-Active)
- Private Hosted Zones & Resolver (VPC DNS, Hybrid Resolution, Forwarding Rules)

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation.

✅ Lab 6: AWS Cost Explorer & Budgets (Set up cost monitoring & budget alerts)

✅ Lab 7: Route 53 DNS Configuration (Create hosted zones, configure A/CNAME records)

✅ Lab 8: Route 53 Failover & Health Checks (Set up active-passive failover with health checks)

Stage 2: Days 27 - 35

Days 27 to 35 will be focusing on **AWS Advanced - Identity and Access Management and Database Services**

Day 27- 35

The following are the Scope of the training for Week 6 & 7

- AWS Identity and Access Management (IAM) concepts (Including users, groups, roles and policies)
- Multifactor Authentication (MFA), Certificate Manager, KMS
- Amazon Relational Database Service (RDS) concepts (Including database instances, security groups, and parameter and option groups)
- Amazon DynamoDB concepts (Including data model and supported operations)
- Shell Script basics

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation.

✅ Lab 9: IAM User & Role Management (Create users, groups, policies & MFA)

✅ Lab 10: Launch & Secure RDS Database (Create RDS instance with security groups & backups)

Continuous Learning: Technical Enablement

Learn the AWS Identity Access Management through the below Udemy course

[AWS Identity Access Management (IAM) Practical Applications]

Learn the Linux for Cloud & DevOps Engineers through the below Udemy course

[Linux for Cloud & DevOps Engineers]

GenAI (Day 21)

Continuous Learning: Technical Enablement

Program completion criteria

Everyone must register for the following e-learning course on **C-Learn** and complete a KBA assessment on Moodle to successfully finish this program.

Online learning: C-Learn

Activity Code: **ELRNG01863**

Fundamentals of Generative AI [101-Basics]

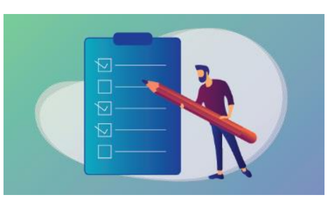
This AI course is designed to equip learners with:

- The foundational knowledge and skills required to harness the power of Generative AI.
- The ability to identify opportunities for innovation and implementation of AI within their organizations.
- The skills to drive organizations toward a future of enhanced creativity and competitive advantage using AI techniques.

GenAI (Day 24)

Continuous Learning: Technical Enablement

Knowledge check - Moodle



Activity Code: **ATHDW335105**

GENERATIVE AI QUICK ASSESSMENT FOR ELEARNING QUIZ [101-BASICS]

- This assessment is to assess the knowledge of associates on Generative AI tools and concepts at a Beginner proficiency.

Stage 2: Days 36 - 41

Days 36 - 41 will be focusing on **Automation**

Day 36 - 41

The following are the Scope of the training

- Automation Build tools(Terraform)
- Python Basics

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation

✅ Lab 15: Setting up Kubernetes environment and deploying a Simple Web App on k8s
✅ Lab 16: Setting up Kubernetes environment and deploying a Simple Web App on k8s Deploying a Canary Deployment with Horizontal Scaling Deploying a Canary Deployment with Horizontal Scaling

Learn the Terraform through the below Udemy course

AWS Infrastructure as Code With Terraform [2024 Updates]

Learn and attend the below Terraform Quiz through Cognizant learn.
TERRAFORM- QUIZ [101-BASICS]

Day 41

Knowledge Based Assessment on AWS Cost Optimization and DNS management, Identity and Access Management and Database Services , Other AWS PaaS Services and Git &Version Control

Note: On your day 42 & 43, you will be taking up interim Technical evaluation by BU SME. You are expected to get RAG Status as “Green” and you have only one attempt to attend interview with BU SME

Evaluation:
Interim Technical Evaluation

Stage 2: Days 44 - 49

Days 44 - 49 will be focusing on **AWS Advanced - Other AWS PaaS Services & Git & Version Control**

Day 44- 49

The following are the Scope of the training

- SNS, SQS, SES(Concept of Building Loosely Coupled Architecture
- Lambda and API Gateway
- CloudTrail, Config, and Trusted Advisor
- Cloud Formation
- Git & Version Control
- Git Basics
- What is Git?
- Installing Git
- Basic Git Commands (init, add, commit, status, log)
- Git Branching & Merging
- Creating & Merging Branches
- Resolving Merge Conflicts
- Git Rebase vs Merge

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation.

✅ Lab 11: SNS, SQS, and SES Integration (Send notifications, trigger workflows, send emails)

✅ Lab 12: Lambda Basics (Write a basic Lambda function & execute it)

☑ Lab 13: Basic Git commands by tracking and pushing changes to GitHub.

Continuous Learning: Technical Enablement

Learn the AWS Advanced - Other AWS PaaS Services through the below Udemy course

AWS Cloud Development Kit - From Beginner to Professional

Learn the GIT & Version control through the below Udemy course

Git Complete: The definitive, step-by-step guide https://cognizant.udemy.com/course/git-complete/to Git

Stage 2: Days 50 - 57

Days 50 to 57 will be focusing on **Docker Essentials & Kubernetes Overview**

Day 50 to 57

The following are the Scope of the training

- Introduction to Docker
- What is Docker & Why Use It?
- Containers vs Virtual Machines
- Installing Docker on Windows/Linux/Mac
- Docker Images & Containers
- Writing Dockerfiles
- Building & Managing Images
- Running, Stopping, Restarting Containers
- Networking in Docker
- Docker Compose & Multi-Container Apps
- Introduction to Docker Compose
- Running Multi-Container Applications
- Docker Volumes & Data Persistence
- Docker Registries & Security Best Practices
- Pushing & Pulling Images from Docker Hub & AWS ECR
- Scanning Images for Vulnerabilities
- Kubernetes Basics & Architecture
- What is Kubernetes?

- Kubernetes Components (Nodes, Pods, Deployments, Services)
- Running a Simple Pod & Service
- Scaling & Updating Applications
- Exposing Applications (ClusterIP, NodePort, LoadBalancer)
- Persistent Volumes (PVs) & Persistent Volume Claims (PVCs)

Continuous Learning: Technical Hands-on

Mandatory Hands-on

Please complete the below mandatory Technical Hands – on and upload the screenshot of your final lab completion in the Teckstac platform and submit for evaluation

✅ Lab 14: Setting up Docker Envionment and deploying simple Web App on Docker.

✅ Lab 15: Setting up Kubernetes environment and deploying a Simple Web App on k8s

✅ Lab 16: Setting up Kubernetes environment and deploying a Simple Web App on k8s Deploying a Canary Deployment with Horizontal Scaling Deploying a Canary Deployment with Horizontal Scaling

Continuous Learning: Technical Enablement

Learn and perform the hands on Docker through the below Udemy course

[Docker for the Absolute Beginner - Hands On - DevOps]

Learn the Continuous Delivery on Kubernetes through the below Udemy course

[Kubernetes for the Absolute Beginners - Hands-on]

Day 57

Knowledge Based Assessment on AWS Advanced modules

Day 58 to 60

Evaluation:
Final Technical Evaluation

Note: On your day 58 to 60, you will be taking up Final Technical evaluation by BU SME

Note: Your interim and final technical evaluation will considered as Gating criteria

What is Final Evaluation?

The Final Evaluation will be conducted to certify whether a GenC is eligible to enter into the BU or not. The skill of a GenC will be gauged on the application development and overall technical knowhow towards the end of GenC Training.

Tech SME from BU will be conducting the final tech evaluation. As a fallback, the project mentor can also steer this activity.

The final evaluation will be conducted as two phases. They are the following

1. **Final Technical Evaluation**
2. **Final Project Evaluation**

The mode of these evaluations will be any one of the following:

- F2F(face to face)
- Video Based

1. Final Technical Evaluation (FTE)

The BU Mentor will interview the GenC on various skills achieved throughout the training program and put a score which will be considered for the final PHS of the GenC.

2. Final Project Evaluation (FPE)

In this evaluation, the BU Mentor will be verifying the skills of a GenC on a project perspective. End of this evaluation, the BU Mentor will score the GenC's work based on various evaluation criterions.

How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Udemy courses and by completing the hands-on exercises mentioned in the daily plan.

The below strategies will help you decide the learning approach

Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

Profile #1

Harry Reacher

Engineering Discipline: Electronics

Skills: Python, Ruby on Rails, nginx

Project: Mining Crime Data to get Route Cause Insights

Learning Approach to Programming Languages: I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

What will work for me?

- Directly complete hands on exercises
- Refer Internet or Udemy Courses
- If hands on are implemented early, clarify your friends questions and troubleshoot their issues

Profile #2

Olivia Richards

Engineering Discipline: Computer Science

Skills: Java, C, C++

Project: Library Management System

Learning Approach to Programming Languages: I have interest, but I don't know where to start.

What will work for me?

- Go through the recommended Udemy Course
- Try completing the hands on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

Profile #3

Greg Anderson

Engineering Discipline: Civil

Skills: C

Project: Fiber reinforced concrete

Learning Approach to Programming Languages: I am scared of programming languages. I haven't got my hands dirty with coding

What will work for me?

- Go through the recommended Udemy Course
- Implement the coding along with the author of the Udemy Course
- Try completing the hands on exercises
- Clarify queries with SME
- Troubleshoot programming issues with help from SME or learner from your classroom whom had already completed

FAQ

1. Who can participate in this program?

   Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GEN Cs can participate in this program.

2. Is there any pre-learning I should do?

   No. This program is open to all students from any academic discipline.

3. How will I know my RAG status?
   Evaluation result will be shared with Candidates by coach.

4. Whom do I reach out in case of any queries?
   Coach is your point of contact.

5. What if I fail in the Interim evaluation?
   Your coach will notify your performance in the Interim evaluation. However, you can continue with the learning.

6. How many chances will I get in the Final evaluation?
   You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.